Software Requirements Specification

for

Workflow Automation System

Version #5

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Revision History

Name	Date	Reason for Changes	Version
Kuba Kałużniak	10.03.2026	Initial version	V1.0
Kuba Kałużniak	17.03.2026	Added references and use case diagrams	V2.0
Kuba Kałużniak	24.03.2026	Added UML class diagrams	V3.0
Kuba Kałużniak	14.04.2026	Added UML state, activity, and sequence diagrams. Modified use case diagram.	V4.0
Kuba Kałużniak	28.04.2026	Deleted template relics and filled out Interfaces section.	V5.0

1. Introduction

1.1. Purpose

The purpose of this document is to present a detailed description of the workflow automation platform. It will describe the usage and features of the system, operations the system is responsible for, and the constraints under which it must operate.

1.2. Document Conventions

There is no significance in the fonts or highlighting used in this specification. Nevertheless, priorities corresponding to a system's feature must be respected by each requirement listed within the feature.

1.3. Intended Audience and Reading Suggestions

The audience shall include company employees, as it is an internal tool intended to be used solely for professional purposes. The system shall be monitored by administrators and supervised by company managers.

The remaining sections of the document provide an overview of the system's implementation, the requirements it shall follow, and characteristics of the users of the product. Section 2 includes general information about the system, including the overall description, its features, type of users and their roles, and operational constraints. Section 3 presents a detailed summary of the system's features. The following sections serve as an extension of Section 3, including functional, non-functional, and interface requirements.

1.4. Product Scope

The system consists of a web-based platform compatible with the company's network and structure. The product's purpose includes optimisation of the company's workflow, leading to dynamic role changes. The system is expected to provide the answer for the challenge of workforce shortage and allow for contracting experts to establish the company's position on the market.

1.5. References

Below is a list of documents which this specification refers to.

Kuba Kałużniak, "Use case diagram examples", last modified 28.04.2026:
https://github.com/kaluzniak34/SE_labs/blob/main/Labs/Lab2_Use_Case_Diagrams/Lab2.pdf

Kuba Kałużniak, "Class diagram examples", last modified 28.04.2026:
https://github.com/kaluzniak34/SE_labs/blob/main/Labs/Lab3_Class_Diagrams/Lab3.pdf

Kuba Kałużniak, "Behavioural diagram examples", last modified 28.04.2026:
https://github.com/kaluzniak34/SE_labs/blob/main/Labs/Lab4_Behavioral_Diagrams/Lab4.pdf

2. Overall Description

2.1. Product Perspective

The product is a web-based platform which integrates with mailing systems, databases, and APIs. It runs on cloud infrastructure. It serves as a replacement for the formerly used automation system, which was deemed obsolete because of the lack of administrative support and limited applications. This product was designed based on the structure of systems from other areas, mentioned in 1.5. References. Then, a use case diagram was designed to outline the tasks each user is assigned to.

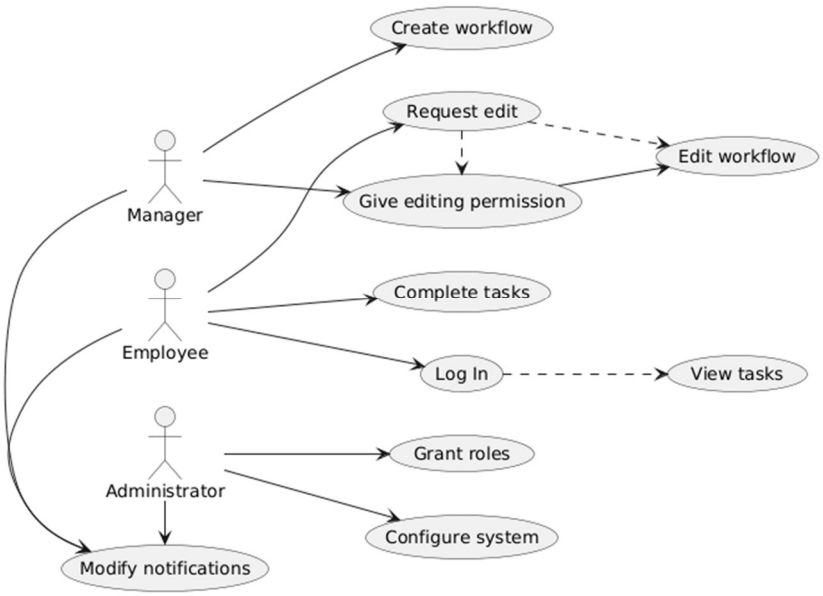

Figure 1. Simplified Use Case diagram of the system.

2.2. Product Features

- Workflow creation and editing
- Task automation
- Notification system
- Dashboard for workflow monitoring
- Integration with external services

2.3. User Classes and Characteristics

User Type	Description
Administrator	Configures the system and integrations
Manager	Creates workflows and monitors performance
Employee	Executes assigned tasks

2.4. Operating Environment

Web browser: Firefox, Chrome, Edge
Operating System: Windows
Applications and technologies used: Fast API, PostgreSQL, JWT

2.5. Design and Implementation Constraints

The system design and implementation are subject to the following constraints:

Technology stack: The system will be implemented using FastAPI for the backend and PostgreSQL for data storage.
Architecture: The system must follow a RESTful API architecture to ensure interoperability with external systems.
Security: Authentication and authorization must be implemented using JWT-based mechanisms.
Performance constraint: The system should support concurrent workflow executions without significant degradation.
Development: The project is developed within limited time and resources.
Browser compatibility: The web interface must support modern browsers.

2.6. Assumptions and Dependencies

Assumptions:

Users have basic knowledge of web applications.
The system will be accessed via a stable internet connection.
Users will operate the system through supported modern web browsers.
Workflows will consist of well-defined tasks.
Authentication tokens (JWT) are securely stored on the client side.

Dependencies:

The system depends on external libraries such as FastAPI and SQLAlchemy.
The system relies on a running instance of PostgreSQL for data persistence.

3. System Features

3.1. Registration

The system grants roles and responsibilities for concurrent users based on their role in the company. The users enter the platform by logging into the organisation's portal, where redirection to the platform occurs.

3.2. Workflow management

The system provides tools which facilitate creating and editing workflows. Managers assess roles and responsibilities to employees, outline goals and set deadlines. Employees, under manager's permission, may perform changes in the workflow.

3.3. Notification system

The product assures users are notified upon a change in the system. Users are able to modify their notification preferences. Notification messages follow a hierarchy, with the highest priority messages being unable to discard.

4. External Interface Requirements

4.1. User Interfaces Overview

The system provides a web-based user interface accessible through modern web browsers. The main interface components include:

- Login Page: Allows users to authenticate using their credentials.
- Dashboard: Displays an overview of workflows, recent executions, and system status.
- Workflow Management Interface: Create, edit, and delete workflows. Define tasks and execution order
- User Management Panel (Admin only): Manage users and roles.
- Notification Panel: Displays alerts and execution results.

The interface will be designed for usability and responsiveness, ensuring compatibility across desktop devices.

5. System Features/Modules

5.1. Registration

5.1.1 Description and Priority

Users can register to the system using their company data. Highest priority.

5.1.2 Stimulus/Response Sequences

Stimulus: Opening the system's web page while connected to the intranet.

Response: Registration or login form is displayed.

Stimulus: Filling in the form with company's data.

Response: Redirecting to the dashboard.

5.1.3 Functional Requirements

REQ-1.1: Upon entering the client-side page, the system shall provide the credential form for the employee.

REQ-1.2: Upon registering, the system shall display the employee's profile with assigned roles and tasks.

REQ-1.3: The system shall grant appropriate privileges to the user based on their role.

5.2. Workflow Management

5.2.1 Description and Priority

Users can organise their work by creating and editing workflows. High priority.

5.2.2 Stimulus/Response Sequences

Stimulus: Manager enters the workflow panel.

Response: Checking if a user is permitted to create a new workflow.

Stimulus: Manager successfully creates a new workflow.

Response: Message is sent to organisational mail addresses of employees involved in the workflow.

Stimulus: Employee asks for editing permissions.

Response: Message is sent to the appropriate workflow manager.

5.2.3 Functional Requirements

REQ-2.1: The system shall allow only managers to create workflows.

REQ-2.2: Upon creating a workflow, the system shall notify the users concerned.

REQ-2.3: The system shall require manager's permission for giving editing privileges to an employee.

5.2.4 Class Diagram

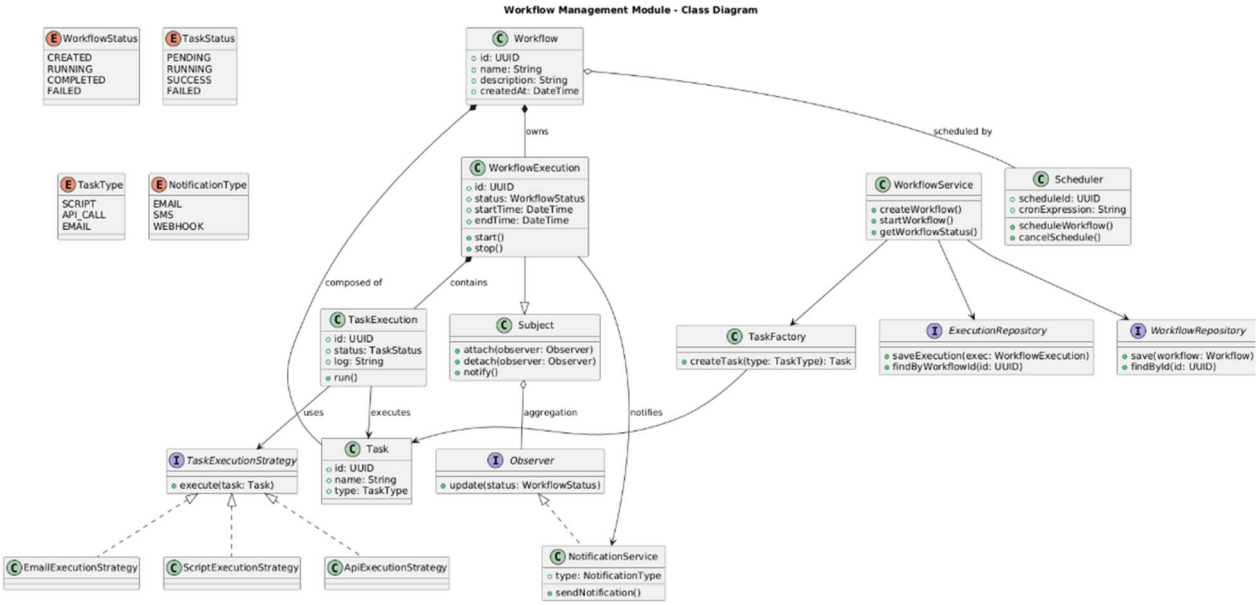

Figure 2. Workflow management module - class diagram.

The class diagram distinguishes between composition and aggregation relationships to accurately represent ownership and lifecycle dependencies. Core entities such as Workflow, Task, and WorkflowExecution are connected through composition, indicating strong ownership. In contrast, components like Scheduler and NotificationService are linked via aggregation or association, reflecting their reusable and loosely coupled nature within the system architecture.

5.2.5 Other Diagrams

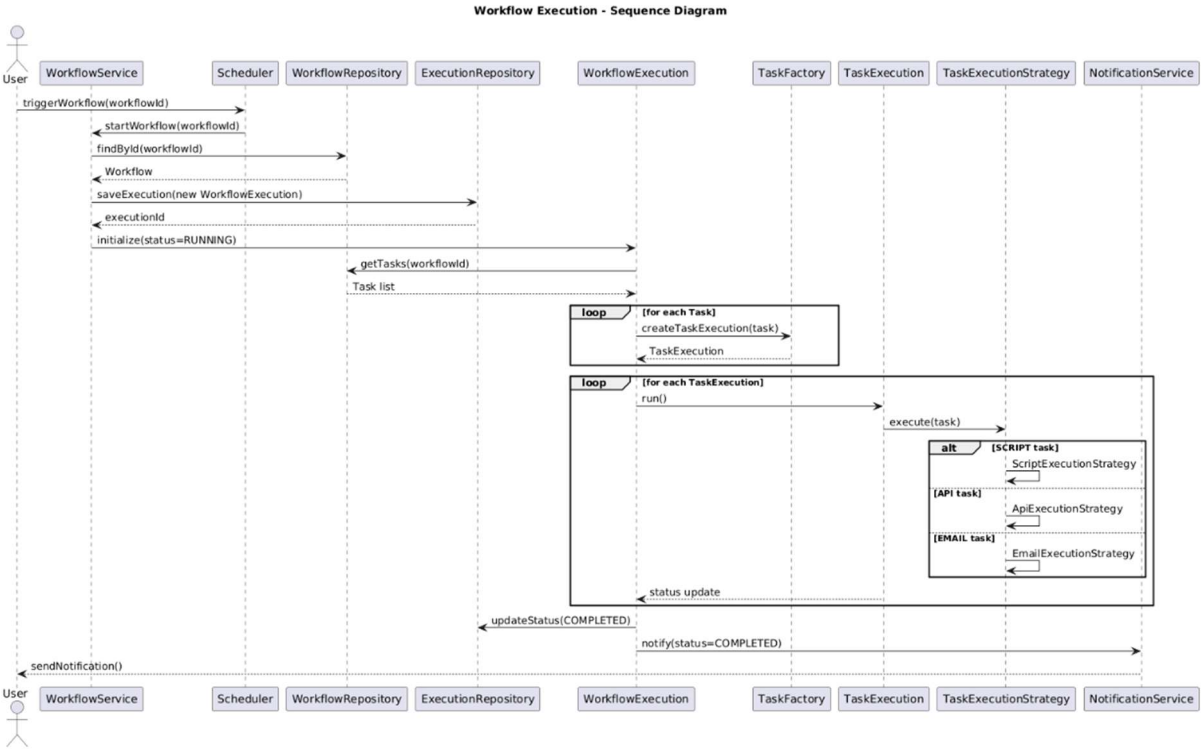

Figure 3. Workflow execution sequence diagram.

The sequence diagram illustrates the runtime interaction between system components during workflow execution. It highlights the orchestration role of WorkflowService, the use of design patterns such as Factory and Strategy for task handling, and the Observer-based notification mechanism upon completion.

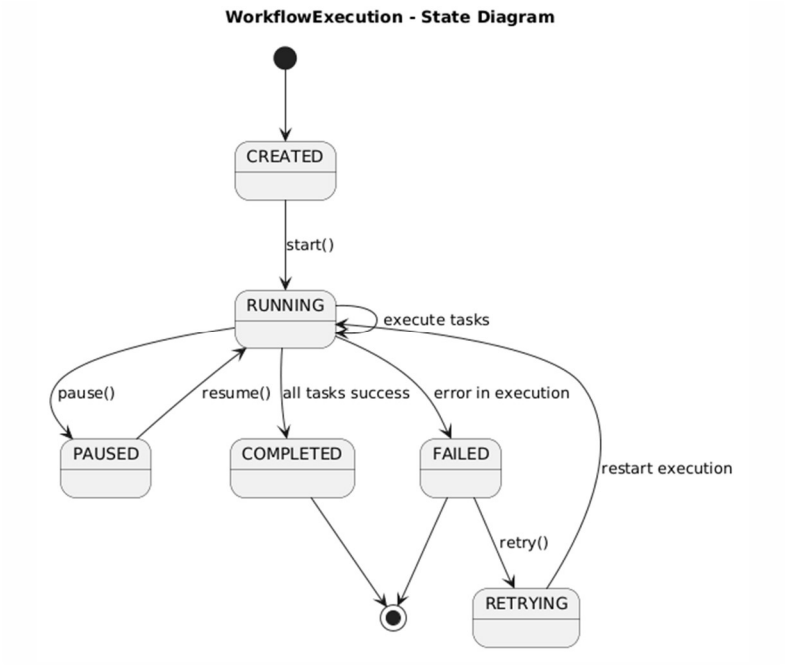

Figure 4. State Diagram – WorkflowExecution Lifecycle.

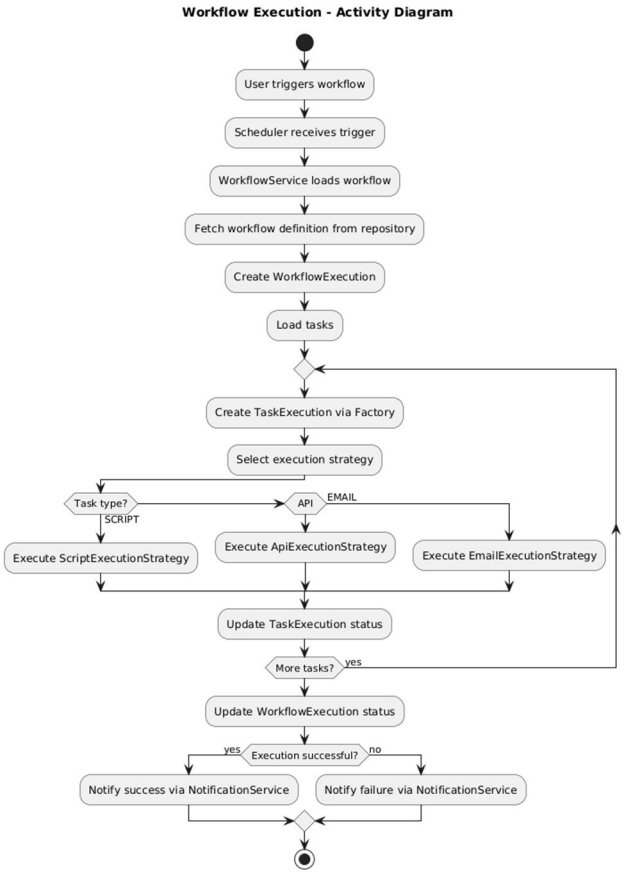

Figure 5. Activity Diagram – Workflow Execution Process.

5.3. Notification system

5.3.1 Description and Priority

Users can modify their notification settings. Notifications follow an importance hierarchy. Low priority.

5.3.2 Stimulus/Response Sequences

Stimulus: User opens the notification settings tab.

Response: A window with various types of notification preferences is displayed.

Stimulus: User confirms the updated notification settings.

Response: Server updates information about user's notification preferences.

5.3.3 Functional Requirements

REQ-3.1: The system shall maintain information about each user's notification settings.

REQ-3.2: The system shall provide a notification importance hierarchy.

REQ-3.3: The system shall disallow turning off notifications with highest importance.

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The system shall support 1000 concurrent workflows.

NF-1.2: The system shall provide response times < 1.5 s.

6.2. Security

NF-2.1: The system shall utilise role-based access control.

NF-2.2: The system shall encrypt all data.

NF-2.3: The system shall be accessed only by the devices connected to the company's intranet.

6.3. Reliability

NF-3.1: The system shall maintain 99.5% uptime.

6.4. Portability

NF-4.1: The system shall support mobile device use.